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A dynamic thresholds scheme for contaminant event detection in water distribution systems

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ABSTRACT

In this study, a dynamic thresholds scheme is developed and demonstrated for contamination event detection in water distribution systems. The developed methodology is based on a recently published article of the authors (Perelman et al., 2012). Event detection in water supply systems is aimed at disclosing abnormal hydraulic or water quality events by exploring the time series behavior of routine hydraulic (e.g., flow, pressure) and water quality measurements (e.g., residual chlorine, pH, turbidity). While event detection raises alerts to the possibility of an event occurrence, it does not relate to origins, thus an event may be hydraulically-driven, as a consequence of problems like sudden leakages or pump/pipe malfunctions. Most events, however, are related to deliberate, accidental, or natural contamination intrusions. The developed methodology herein is based on off-line and on-line stages. During the off-line stage, a genetic algorithm (GA) is utilized for tuning five decision variables: positive and negative filters, positive and negative dynamic thresholds, and window size. During the on-line stage, a recursively Bayes' rule is invoked, employing the five decision variables, for real time on-line event detection. Using the same database, the proposed methodology is compared to Perelman et al. (2012), showing considerably improved detection ability. Metadata and the computer code are provided as Supplementary material.

1. Introduction

Incidents of contaminant injections into water distribution systems emphasize the need for contamination event detection systems. Events like the poisoning of drinking water in Scotland (Gavriel et al., 1998) and the water supply system poisoning in Japan (Yokoyama, 2007) highlight how intentional sabotage remains a major concern to public health (WHO, 2004; Greenfield et al., 2002). Accurate alerts in real-time are required with respect to response strategies, operation optimization, and overall system efficiency (Kroll, 2006; Jain and McLean, 2006; Hasan et al., 2004).

Contamination event detection methods are aimed at identifying abnormal system behavior. Identification of most contaminants typically requires grab sampling followed by laboratory analysis (ASCE, 2004). Since on-line instrumentation capable of measuring and detecting all possible contaminants does not exist, the presence of pollutants can be inferred only through surrogate measurements. Such measurements can distinguish irregularities in monitored water quality, hydraulic parameters, and their interplay during normal operating conditions. This approach rests on the premise that a contaminant injected into a water distribution system (WDS), whether deliberately, accidentally, or

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naturally, will affect at least one of the on-line monitored hydraulic and/or water quality parameters and the interactions between them. Based on this notion, the analysis of continuously monitored data can be utilized to give an indication of contaminants' presence in a water distribution system.

The US government generated research in this field by establishing the Environmental Technology Verification program in 1995 to investigate how changes in water quality parameters can be detected by real-time sensors. Extensive research is being performed by academics, governmental agencies, private companies, and in close collaborations between them. These projects range from theoretical to applied research on commercially-developed hardware and software. Klise and McKenna (2006) and McKenna et al. (2008) suggested three approaches to water quality detection based on a comparison between predicted and measured values that evaluate each technique through receiver operating characteristic (ROC) curves. The US Environmental Protection Agency (EPA) has conducted experiments on more than 30 contaminants (e.g., pesticides, insecticides, metals, bacteria), which may be employed in intentional acts of water contamination (U.S. EPA, 2005a; U.S. EPA, 2005b). Hall et al. (2007) provided results from experiments that tested the response of commercial water quality sensors of different designs and technologies to chemical and biological loads. These tests suggested that free chlorine, total organic carbon (TOC), oxidation reduction potential (ORP), conductivity, and chloride were the most reactive parameters to the majority of contaminants. Yang et al. (2009) tested 11 contaminants at different concentrations using a pilot-scale pipe system. The suggested adaptive transformation of sensory measurements reduced background noise and enhanced contaminant signals. This allowed for contaminant detection and further classification based on chlorine kinetics.

Today, water utilities have access to a variety of private and public event detection hardware and software. Hach (GuardianBlue™) produces commercial hardware and software solutions for event detection. Edthofer et al. (2010) provides event detection software combined with management and data validation tools. The CANARY software (Hart et al., 2007, 2010; Murray et al., 2010), developed at Sandia National Laboratories in collaboration with the EPA National Homeland Security Research Center, is a freely available tool for detecting contamination events. It provides both off-line and on-line analysis capabilities for detecting anomalies in regularly-monitored water quality and hydraulic parameters that indicate possible contamination events.

The development of multi-parameter water quality models and their calibration is challenging due to the large amount of information and number of parameters involved (e.g., Chungsying et al., 1994; Rauch et al., 1999). Thus, their application in the context of contamination event detection is complicated.

Data-driven models such as artificial neural networks (ANNs) have been successfully used for water quality assessment that focuses on modeling and prediction. This includes prediction of residual chlorine, the temporal variation of substrate, biomass concentrations, and residual chlorine (e.g., D'Souza and Kumar, 2009, Gibbs et al., 2006; Rodriguez et al., 2002; Rodriguez and Serodes, 1999). Arad et al. (2011) utilized decision trees for evaluating event detection procedures. Later, Perelman et al. (2012) utilized ANN to improve the performance of the estimation model. To detect possible quality threats in water distribution systems, Perelman et al. (2012) integrated ANN models with Bayesian sequential analysis to estimate the probability of such events. Closely related methodologies were also utilized to detect leaks and bursts in water distribution systems (Romano et al., 2010a, 2010b).

The method described below attempts to improve the study of Perelman et al. (2012) by applying adaptive updating dynamic thresholds, which are used to classify sensory observations. Dynamic, as opposed to fixed, thresholds are characterized by a sliding window size, and positive and negative dynamic thresholds that are computed as a function of the error standard deviation of the sliding window. In addition, dynamic thresholds are characterized by and positive and negative filters, which discriminate possible outliers in the measurements.

In this method, optimal parameter values are obtained using the commonly applied genetic algorithm (GA) (Holland, 1975; Goldberg, 1989). The following sections provide details on the suggested method and its application to contamination event detection in water distribution systems.

2. Methodology

The general event detection framework relies on continuously transmitted hydraulic and water quality data from the supervisory control and data acquisition (SCADA) system.

A previously published event detection method (Perelman et al., 2012) consists of four main modules for each incoming observation from the SCADA system: (1) data analysis – the interplay between water quality parameters are examined using ANNs, (2) identification of outliers – calculated residuals are classified as normal or abnormal observations using fixed thresholds for each parameter, (3) identification of events – based on consequent classification of errors, events are identified by updating Bayesian probability, and (4) synchronized decision – information from multiple quality indicators is fused to provide a unified decision support about a contamination event. Steps 1 and 2 are initially trained using available data collected from the water utility. Steps 1–4 are then repeatedly executed in real-time for each new observation. This study suggests improving stage 2 by identifying outliers using an updating dynamic threshold instead of a fixed one. This would upgrade all subsequent stages/events identification and fused decisions. A summary of the overall event detection methodology is presented in Fig. 1 and described in following sections.

2.1. *Error calculation*

The input parameters attained from the SCADA system include hydraulic (pressure) and water quality (e.g., residual chlorines, pH) time series data.

Artificial neural networks (ANNs) are utilized to estimate the relationships between water quality parameters during normal operation. The development of ANN does not require

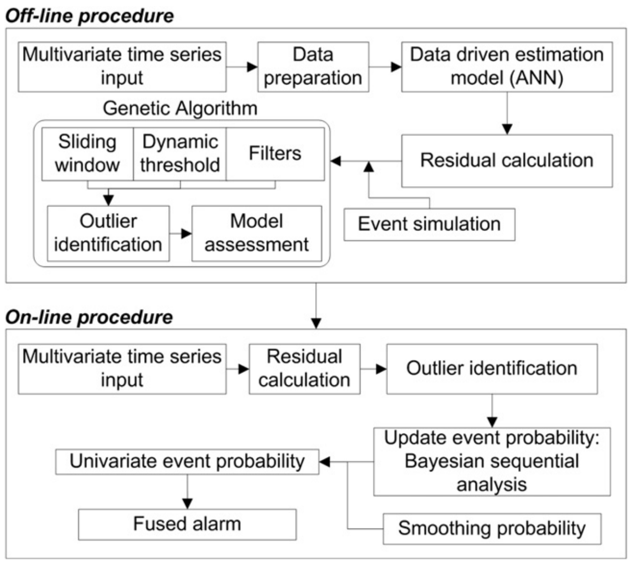

Fig. 1 – Methodology flow chart.

any knowledge of the physical and chemical laws governing the water quality parameters. This makes them an attractive tool for modeling complicated multivariate reactions in water supply systems.

For each target water quality parameter, an ANN model is constructed with the input containing measured time series of all predictive parameters and lagged target parameters, as described in Eq. (1):

$$\widehat{x}_{i,t} = f(x_{1,t}, ..., x_{i-1,t}, x_{i,t-1}, x_{i+1,t}, ..., x_{n,t}) \quad (1)$$

where $x_{i,t}$ and $\widehat{x}_{i,t}$ are the measured and estimated water parameter i at time t, respectively, n is the number of water quality parameters, and $f(\cdot)$is a function attained by the ANN estimation model.

The estimated values are then used to calculate the residual, defined as the difference between the measured and estimated values, as shown in Eq. (2).

$$Y_{i,t} = x_{i,t} - f_i(\bullet) = x_{i,t} - \widehat{x}_{i,t} \quad (2)$$

where $Y_{i,t}$ is the estimated residual for parameter i at time t.

2.2. *Event simulation*

For obvious reasons, the behavior of contaminants and their effect on water parameters in a real water distribution system cannot be tested. There are two methods for performing simulations of contamination events: (1) pilot-scale laboratory tests, and (2) computer-generated events.

Testing contaminants in laboratories may be limited as there are numerous contaminants, each of which behaves differently in every water distribution system due to quality differences, demand fluctuations, hydraulic operations and other unique characteristics. The use of computer codes to simulate contamination events can overcome these difficulties by introducing generic disturbances to the measured time-series data. Although these simulations do not always reflect the true chemical and biological underlying processes, this approach can be justified as the proposed method assumes that standard measured water quality parameters will react in the presence of some contaminants. The developed event detection method should thereafter be tested for a wide range of variables on an experimental setup. The simulated events are imposed on routine data and represent the effects of a contamination event on the system. These events may vary in magnitude, direction, length and water quality parameters. Details about simulation of contamination events in this work are given in the application section.

2.3. *Outlier identification*

Fixed thresholds (Perelman et al., 2012) and the new dynamic thresholds approach for outlier identification are presented here.

2.3.1. *Fixed thresholds*

To distinguish the system's normal operation conditions from contamination events, the computed residuals $Y_{i,t}$ are classified as "Normal" or "Outlier". Previous work (Perelman et al., 2012) employed constant thresholds to classify errors. Values bounded within the thresholds, which remained constant, were considered to represent normal operation conditions. The threshold values were determined based on ANN errors during normal operation without events. This meant that the majority of residual (between 96% and 99%) resided within the upper and lower limits (Perelman et al., 2012).

2.3.2. *Dynamic thresholds*

This study develops a dynamic threshold method aimed at improving the results of the fixed thresholds method in the outliers' classification module of the methodology. This, in turn, will improve the event detection results.

Dynamic, as opposed to fixed, thresholds are characterized by five parameters: sliding window size (w), positive and negative dynamic thresholds (PDT, NDT) computed as a function of the error standard deviation of the sliding window, and positive and negative filters (PF, NF), which discriminate possible outliers in the measurements. The genetic algorithm (GA) optimization technique is utilized to seek the optimal parameters of the dynamic threshold. GA, invented by Holland (1975), is a heuristic search technique which imitates the mechanics of natural selection, involving iterative solutions through population generations, selection, crossover, and mutation. GA's are common optimization techniques applied in various engineering fields. The reader is referred to Goldberg (1989) for further details.

2.3.3. *Decision variables*

2.3.3.1. *Window size*. The window size is the number of past observations used to calculate the mean Mean (Y) and standard deviation Std (Y) of the errors to determine the positive and negative dynamic thresholds. For each parameter i and every time step t, a new observation enters the sliding window, the oldest one exits (i.e., first in first out (FIFO)), and the errors' standard deviations are recalculated.

2.3.3.2. *Positive and negative dynamic thresholds*. Positive and negative dynamic thresholds are multipliers of the standard deviation of errors within the window size as shown in Eq. (3). Standard deviation changes with every new observation entering the sliding window. This changes the thresholds in every time step and creates dynamic thresholds.

2.3.3.3. Positive and negative filters. Measured data can include outliers resulting from mechanical faults such as malfunctioning sensors, communication disruptions or other interferences. Calculation of the standard deviation is biased toward these extreme observations and consequently, the GA performance. As a result, filters are deployed. These positive and negative filters determine which observations will be used in the standard deviation calculation in each window and which will be left out. The filters are measured in absolute values and remain fixed throughout the process. It is important to note that these observations are ignored only in the standard deviation calculation process. Once all five decision variables are set, all observations, including those with extreme values, are taken into consideration in the analysis.

The optimal parameters of the dynamic threshold are computed off-line based on historic measured data with imposed simulated events. The attained values are then used to classify errors in real-time according to:

$$\begin{aligned} &Y_{i,t} > \mathrm{Mean}(Y_{i,f,w}) + \mathrm{PDT}_i \times \mathrm{Std}(Y_{i,f,w}) \text{ or} \\ &Y_{i,t} < \mathrm{Mean}(Y_{i,f,w}) - \mathrm{NDT}_i \times \mathrm{Std}(Y_{i,f,w}) \end{aligned} \tag{3}$$

where $Y_{i,t}$ is the ANN error for water quality parameter i at time t, Mean ($Y_{i,f,w}$)and Std ($Y_{i,f,w}$) are the mean and standard deviation of filtered errors of water quality parameter i in the sliding window w, respectively, and $\mathrm{PDT}_i$and NDT_i are the positive and negative dynamic thresholds multipliers, respectively. At each time t, model errors are tested against these bounds and are classified as outliers, if exceeding them.

2.3.4. Objective function

A confusion matrix represents the model's classification of all residuals Y_i to one of four classes: true positive (TP) – the residual was classified as an outlier during an actual contamination event, false positive (FP) – the residual was classified as an outlier during routine operation conditions, true negative (TN) – the residual was classified as a plausible model error during routine operation conditions, and false negative (FN) – the residual was classified as a plausible model error during a contamination event. The GA performance measurement is set to maximize the trace of the confusion matrix, i.e., TP + TN, with w, PDT, NDT, PF, and NF being the decision variables. Additional commonly used metrics, sensitivity and specificity, can be derived from the confusion matrix as described in Eq. (4):

$$\begin{aligned} &\text{Sensitivity} = \text{True positive rate (TPR)} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \\ &1-\text{Specificity} = \text{False positive rate (FPR)} = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}} \end{aligned} \tag{4}$$

Complementary, TPR + FNR = 1 and TNR + FPR = 1, where FNR and TNR are the false and true negative rates, respectively.

2.4. Events identification

To determine the possibility of an event occurrence from previously classified observations ("Normal" or "Outlier"), the probability of a contamination event is updated on-line based on the classification of the residuals. Initially, the probability of a contamination event is assumed to be rare. With each new observation, the probability of an event is sequentially updated using Bayes' rule: 286

$$P(E_t) = \begin{cases} P(E_t|O_t) & \text{if Residual}_t \text{ is Outlier} \\ P(E_t|\overline{O}_t) & \text{if Residual}_t \text{ is Normal} \end{cases}$$

where

$$\begin{aligned} P(E_t|O_t) &= \frac{P(O|E)\times P(E_{t-1})}{P(O|E)\times P(E_{t-1}) + P(O|\overline{E})\times P(\overline{E}_{t-1})} = \frac{\mathrm{TPR}\times P(E_{t-1})}{\mathrm{TPR}\times P(E_{t-1}) + \mathrm{FPR}\times[1-P(E_{t-1})]} \\ P(E_t|\overline{O}_t) &= \frac{P(\overline{O}|E)\times P(E_{t-1})}{P(\overline{O}|E)\times P(E_{t-1}) + P(\overline{O}|\overline{E})\times P(\overline{E}_{t-1})} = \frac{(1-\mathrm{TPR})\times P(E_{t-1})}{(1-\mathrm{TPR})\times P(E_{t-1}) + (1-\mathrm{FPR})\times[1-P(E_{t-1})]} \end{aligned} \tag{5}$$

where $P(E_t)$ is the probability of an event at time t, O_t and $\overline{O}_t$ are "Outlier" and "Normal" observations at time t, respectively, TPR and FPR are true and false positive rates, respectively, and $P(E_t|O_t)$ is the conditional probability of a contamination event given that the residual is classified as an outlier. The TPR and FPR are derived from the objective function of the GA optimization of the dynamic threshold.

Event probability is updated for each parameter independently resulting in univariate event probabilities. The probability of events designates the likelihood of a contamination event based solely on the target parameter. An event is declared when the probability exceeds a threshold value. This value is a customized parameter set by the decision maker, reflecting his attitude toward risk.

2.5. Fused alarms

Univariate event probabilities are fused to give a unified multivariate event probability. The multivariate probability reflects the likelihood of a contamination event based on all data analyzed from all parameters. Since water quality parameters differ in their predictive ability of contamination events, weights reflecting their influence on the synchronized decision should be allocated. In this study, uniform weights were given to reflect no prior information. Other methods to assign weights may utilize receiver operating characteristic (ROC) curve information or expert opinions.

3. Application

Real data collected from a water utility, available from the CANARY database, was utilized. The data spans over four months with 5-min intervals of normal operation conditions, and consists of six water quality parameters: total chlorine (mg/L), electrical conductivity (EC) (mS/cm), pH (–), temperature (°C), total organic carbon (TOC) (ppb), and turbidity (NTU). The original dataset was divided into two subsets. The first, containing 67% of observations (~22,000 time steps), was used to create, train, and evaluate the model. The second dataset, with 33% (~11,000 time steps), was used for testing the model accuracy, resembling real time on-line decision making.

3.1. *Event simulation*

Four types of contamination events, characterized by magnitude and direction, were simulated: high impact events, random impact events, low impact events, and mixed impact events with 70% random and 30% low impact events. The magnitude of impact differs from 2.5 standard deviations, from routine patterns in the high impact events, to 0.5 standard deviations in the low impact events. Events were assumed to be Gaussian, with durations of 8 h. Once generated, the events were superimposed on the collected routine data. Fig. 2 depicts an example of a partial time series plot of the four types of contamination events imposed on routine operations of total chlorine. The event simulation scheme is based on the methodology presented in Klise and McKenna (2006) and further detailed in Perelman et al. (2012).

3.2. *Error calculation*

Six ANN models, one for each water quality parameter, were created and trained to estimate accuracy as compared to the input measured time series presented in Eq. (1). The residuals between the estimated and measured values were calculated according to Eq. (2). A feed-forward back-propagation network with twenty neurons in the hidden layer was constructed using the Neural Network Toolbox of Matlab™ (MATLABTM, 2010). The network was trained with tan-sigmoid transfer function in the hidden layer and linear transfer function in the output layer.

3.3. *Dynamic thresholds*

A GA optimization was used to find the optimal parameters of the dynamic threshold to classify observations (Section 2.3.2). The optimization toolbox of Matlab (MATLABTM, 2010) was used for the GA optimization with a population size of 30 and 50 generations. The lower and the upper boundaries (LB and UB) for the five decision variables w, PDT, NDT, PF, and NF, were set to LB = [0.01, 0, 0, 0, −10] and UB = [0.2, 2, 2, 10, 0], respectively. Fig. 3 illustrates all five GA decision variables and Table 1 details GA results for the window size, positive and negative dynamic threshold, and positive and negative filters of an example run during a random impact simulated event.

It can be seen from Table 1 that the window sizes vary from 8 h, corresponding to the electrical conductivity, to 45 h for the total organic carbon (TOC). These variations illustrate the different behavior of each measured parameter (i.e., mean, standard deviation, and contamination events identification intensity). The size of each sliding window is a demonstrative character of every parameter found by the GA to increase the number of true positives and true negatives.

The positive and negative dynamic thresholds are the multipliers of standard deviation. For example, the negative

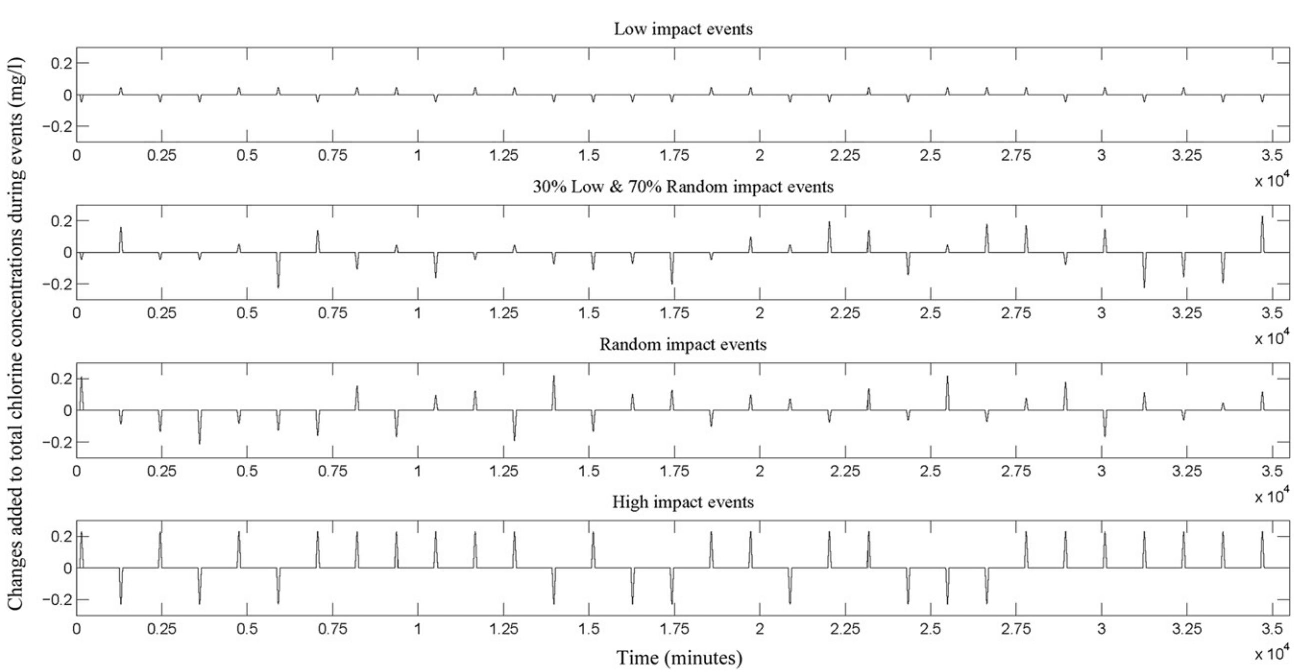

Fig. 2 – Contamination event types.

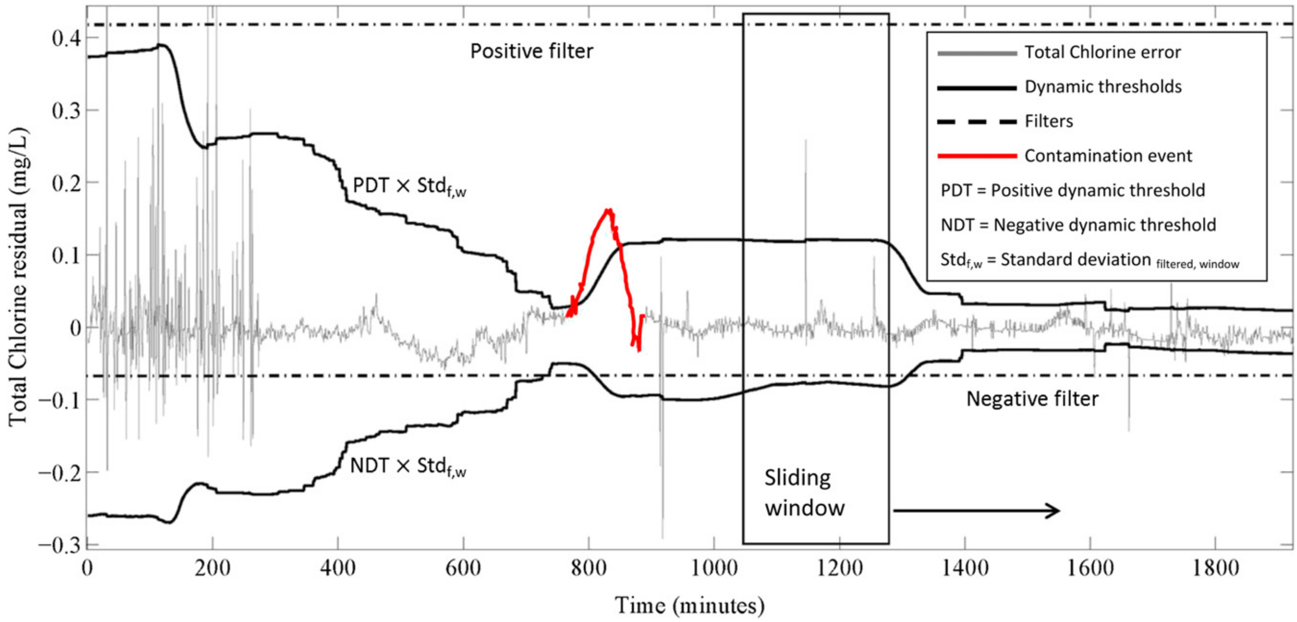

Fig. 3 – Genetic algorithm decision variables: Positive filter, negative filter, PDT, NDT, sliding window size.

dynamic threshold of TOC is 0.667. This value is then multiplied by the standard deviation of the filtered observations in each window. The better the estimation model, the lower positive and negative dynamic thresholds should be. This is attributed to the low standard deviation of the residual in each window. The positive and negative filters are presented as absolute values. These filters are used to screen extreme observations when calculating the standard deviation in each

Table 1 – Genetic algorithm example results.

Parameter	Positive filter	Negative filter	Positive dynamic threshold (PDT)	Negative dynamic threshold (NDT)	Sliding window size (Hours: minutes)
Total chlorine (mg/L)	2.725	−6.454	1.998	1.999	19:36
Electrical conductivity (mS/cm)	9.637	−9.955	1.989	1.978	08:11
pH (–)	1.945	−0.812	1.997	1.982	16:13
Temperature (°C)	3.278	−1.623	1.996	1.945	27:18
Total organic carbon (ppb)	4.667	−8.178	0.970	0.667	44:42
Turbidity (NTU)	0.262	−5.713	1.013	1.500	43:52

Table 2 – Event probability example table.

Time	Monday 08:00	Monday 08:20	Monday 08:40	Monday 09:00	Monday 09:20	/	Monday 11:20	Monday 11:40	/	Monday 15:40	Monday 16:00	Monday 16:20	Monday 16:40	Monday 17:00
True state	False	*True*	*True*	*True*	*True*		*True*	*True*		*True*	*True*	*True*	False	False
Event Probability														
Total chlorine	0.0000129	0.021	0.575	0.95*	0.95	/	0.95	0.95	/	0.572	0.373	0.00367	0.00258	0.000309
Electrical conductivity	0.00001	0.00001	0.00001	0.00001	0.00001	/	0.783	0.90	/	0.95	0.95	0.95	0.861	0.0464
pH	0.00001	0.00001	0.00001	0.00001	0.00001	/	0.521	0.781	/	0.744	0.545	0.00569	0.00395	0.000414
Temperature	0.0000527	0.000034	0.0000219	0.000011	0.00001	/	0.00001	0.00001	/	0.00662	0.00427	0.000198	0.000104	0.000279
Total organic carbon (TOC)	0.00001	0.00001	0.00001	0.0856	0.837	/	0.95	0.95	/	0.95	0.95	0.773	0.462	0.508
Turbidity	0.0000781	0.553	0.926*	0.95	0.95	/	0.95	0.95	/	0.95	0.95	0.00363	0.194	0.00239

/ = time gap, and 0.95* = above threshold probability of 0.95 ($P_{Threshold}$ = 0.7) indicating a contamination event.

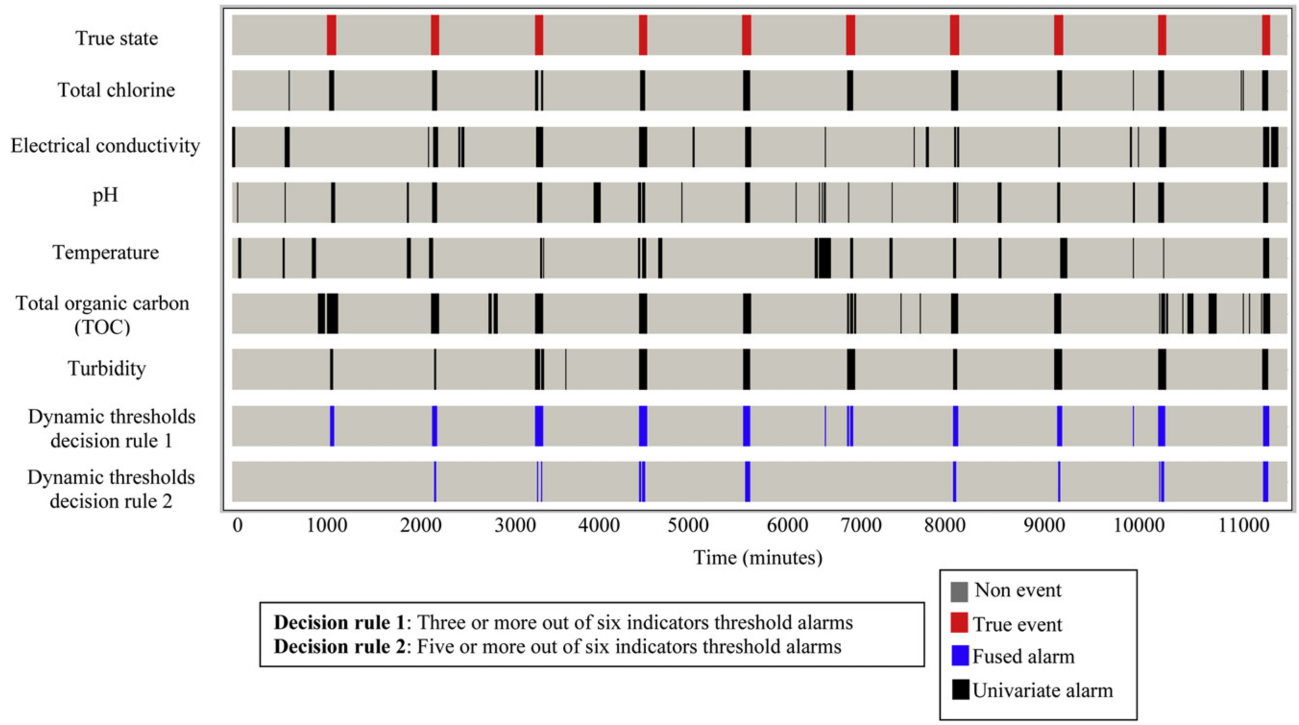

Fig. 4 – Dynamic thresholds univariate and fused alarms.

window of every parameter. When residuals vary only slightly, these filters receive relatively low values (e.g., pH filters). However, when the parameter's residual is erratic, these values are larger (e.g., electrical conductivity).

After training the ANN models and running GA optimization, the attained results are repeatedly evaluated in real-time for each new observation. The ANN models and dynamic threshold parameters need to be periodically updated as new measurements become available, but not on hourly or even daily basis. Running times for creating six ANN models were around 20 min, running GA optimizations to find the optimal parameters for each models (5 decision variables) took

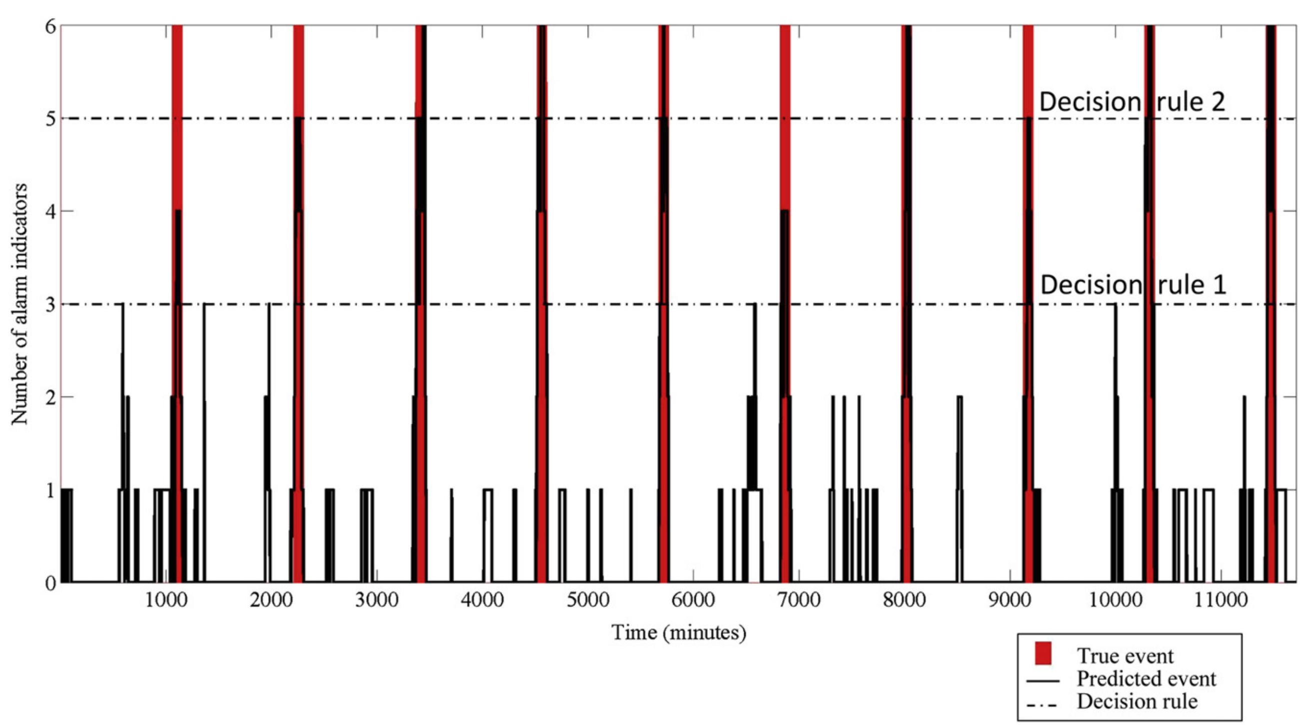

Fig. 5 – Multivariate fused alarms.

Table 3 – Fused alarms comparison to Perelman et al. (2012).

Simulated events	Model	Alarm type	Number of alarm indicators					
			1	2	3	4	5	6
High impact events	FT	TP	10	10	10	10	8	2
		FP	105	5	0	0	0	0
	DT	TP	10	10	10	10	10	7
		FP	58	11	2	0	0	0
Random impact events	FT	TP	10	10	9	7	5	2
		FP	89	13	0	0	0	0
	DT	TP	10	10	10	10	8	6
		FP	74	23	6	0	0	0
Low impact events	FT	TP	10	7	2	2	1	0
		FP	72	2	1	0	0	0
	DT	TP	10	10	10	7	4	2
		FP	55	27	9	1	0	0
30% low and 70% random impact events	FT	TP	10	9	8	6	4	2
		FP	85	14	1	0	0	0
	DT	TP	10	10	10	9	8	5
		FP	73	25	7	1	0	0

FT = fixed thresholds (Perelman et al., 2012), DT = dynamic thresholds (current study), TP = true positive, and FP = false positive.

approximately 40 min, and evaluating each new observation and making a decision in real-time about the probability of an event is instant. All running times are reported on Intel® 8 GB 2.80 GHz.

The observation of each parameter is part of an ongoing process aimed at providing the decision maker with the probability of a contamination event. After each observation is classified as "Normal" or "Outlier", the probability of a contamination event is updated using Bayes rule (Eq. (5)). Table 2 shows an example of a contamination event and its corresponding probabilities as estimated by the model. The alarm of each parameter is raised only when the probability exceeds a predetermined critical probability value. In this work, the threshold was set to $P_{Threshold} = 0.7$. For example, at 09:20, three of the six parameters identified a contamination event, and by 11:40 five of the six raised an alarm.

Fig. 4 illustrates the dynamic threshold univariate and fused alarms. The first row depicts the simulated events imposed on each of the parameters. Each of the six following rows represents one water quality parameter, and the model's estimation of a contamination event based solely on that parameter (black to indicate events and grey to indicate normal operational conditions). While some of the indicators give better estimations than others, it is clear that the number of false alarms is high for each individual parameter.

To improve the model's estimation of an event and to lower the number of false alarms, an information fusion process is invoked. This process ensures that only when a predetermined number of parameters indicate an event, an alarm is raised. This lowers the number of false alarms (false positives) and gives the decision maker confidence in the raised alarms' reliability.

Fig. 5 depicts the multivariate fused alarms. It can be seen from Fig. 5 that when the decision maker sets the alarm to three parameters or more (i.e., decision rule 1), the model identifies all ten contamination events at the expense of five

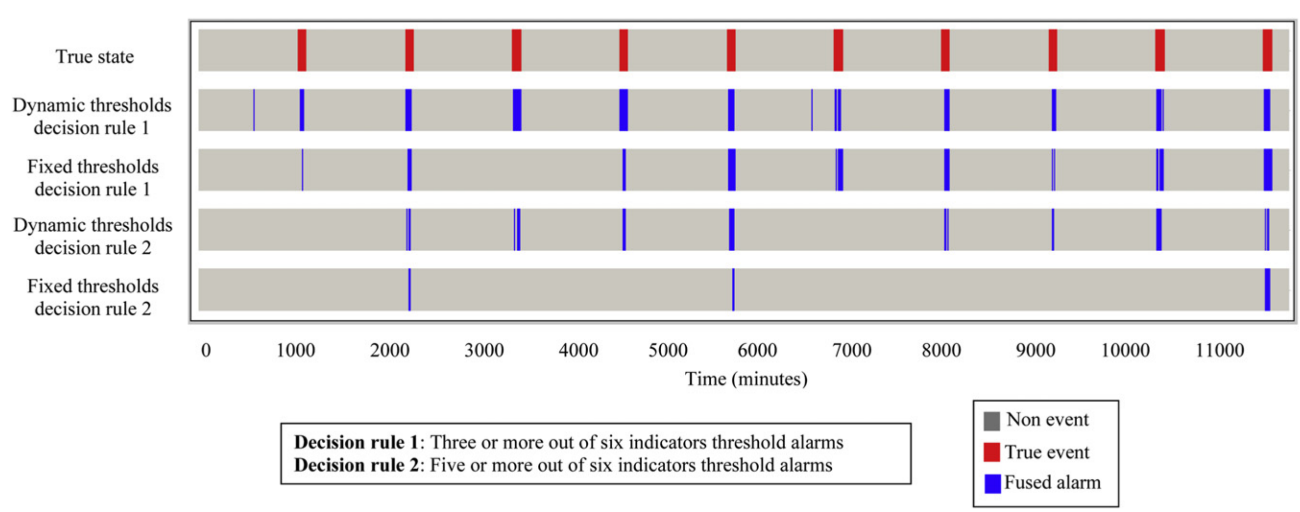

Fig. 6 – Dynamic and fixed thresholds fused alarms comparison.

false alarms. If the decision rule determines that the alarm is raised when five or more parameters indicate an event (i.e., decision rule 2), there are no false alarms and eight out of the ten contamination events are identified. This reflects the basic tradeoff between the sensibility and reliability of the method.

3.4. *Comparison to Perelman et al. (2012)*

The dynamic thresholds method is compared to the fixed thresholds method (Perelman et al., 2012) on identical multiple runs. Table 3 compares the results to Perelman et al. (2012) for four simulated types of events, each with multiple runs. The values in the table represent averaged true positives and false positives of detected contamination events from the ten simulated events.

It can be seen from Table 3 that the dynamic thresholds method detects more events with higher true positives, whereas the fixed thresholds method has lower false alarms. For example, when looking at decision rule 1 (i.e., third column in "Number of alarm indicators") at "30% low and 70% random impact events" type, the dynamic thresholds detected ten out of ten contamination events, while the fixed thresholds identified eight out of those same ten. However, the dynamic threshold method created seven false alarms, in contrast to the fixed threshold's one.

Fig. 6 describes an example random event type run and final results for both decision rules and for both methods. The alarms (true and false) are distributed along the time axis, with their width representing the duration of the identified event. The real events (red) widths are fixed (8 h), and as seen in Fig. 6, the closer the width of the estimated event to the real event, the better the model's estimation. For example, the first event is detected both by the dynamic and the fixed model decision rule 1. However, the width of the estimated event in the dynamic threshold model is closer to the width of the simulated event, illustrating this method's superior detection abilities.

In both Figs. 5 and 6 a delay in the raised alarms compared to the events is visible. This delay is a result of the time required for the Bayesian analysis to increase the probability, based on outliers exceeding the thresholds, until it reaches the critical probability. In all instances, the starting point of an event can be deduced by going back in time to the point the event was detected.

The computer code of the proposed methodology and metadata on the program structure are provided herein as Supplementary material.

4. Conclusions

A contamination event detection method designed to identify anomalous behavior of water quality parameters based on Perelman et al. (2012) was developed and demonstrated. The method utilizes ANN to capture the temporal variations in water quality parameters. Dynamic thresholds differentiate between residuals representing normal operation conditions and contamination events. The parameters of the dynamic thresholds method are sought through a GA formulation, which maximizes the sum of the true positive and true negative indicators. Bayesian sequential analysis transforms the detected outliers to a contamination event probability. A fusion scheme assimilates the contamination event probabilities attained from all water quality parameters into a single integrated comprehensive contamination event probability figure.

The method presented can be implemented in any water distribution system given that proper preparation is made along with settings of some parameters, such as the initial and critical probabilities of a contamination event.

The application demonstrated how the method copes with real data. The technique was tested on different time steps, water quality parameters, and types of contamination events. The methodology can provide both visual and statistical indications of contamination events. These indications are based on single or multiple routinely measured parameters. As the exact behavior of contaminants in real water distribution systems is unknown, contamination events were simulated, yet with different intensity/sever impact characteristics.

The decision rules presented examples for the decision maker's tradeoff in selecting the number of parameters required to induce a contamination event. In the most conservative approach, an event occurrence will be announced if at least one indicator identified such an event. The most lenient approach will demand the presence of all parameters that indicate contamination before an alarm is raised.

Further research is warranted to incorporate more representative simulations of contamination events in water distribution systems, and in integrating multiple sensor information into event detection methodologies. Improvements to the proposed method can also result from better fusion using unequal water quality weights. This is based on their likelihood of reliably triggering a contamination intrusion.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.watres.2013.01.017.

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